

2D Array Interview Practice Problems

This document contains Easy to Hard level 2D array (matrix) problems commonly asked in technical interviews. Try solving each problem within the suggested time limit. No solutions are provided.

Easy Level Problems

Problem 1: Print Matrix Row-Wise — Given a 2D array, print all elements row by row.

Problem 2: Sum of Each Row — Given a 2D array, print the sum of elements of each row.

Medium Level Problems

Problem 3: Search an Element in 2D Array — Given a matrix and a target value, return true if it exists, else false.

Problem 4: Find Largest Element in Each Column — For every column, find the maximum element.

Problem 5: Count Even and Odd Numbers — Count how many even and odd numbers are present in the matrix.

Problem 6: Diagonal Sum of Matrix — Given a square matrix, find the sum of primary and secondary diagonals. If an element overlaps, count it only once.

Problem 7: Row with Maximum Sum — Find the row index that has the maximum sum of elements.

Medium–Hard to Hard Problems

Problem 8: Transpose of a Matrix — Convert the given matrix into its transpose.

Problem 9: Spiral Matrix Traversal — Print the elements of a matrix in spiral order.

Problem 10: Search in a Sorted Matrix — Given a matrix where each row and column is sorted, search for a target value efficiently.

Recommendation: Attempt problems in increasing difficulty order and strictly follow the time limits to simulate interview conditions.